The Longitudinal Wave Problem in the Elastic Space Paradigm

Zou, Zhi Kai 邹志凯 Independent Researcher

Email:zhiyan.zou@foxmail.com Wuhan; China

https://orcid.org/0009-0000-4279-1064

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Abstract

Since Clifford proposed the conception that "matter is curvature of space" in the 19th century, the elastic space (elastic spacetime) paradigm has exhibited strong ontological appeal. It offers an intuitive path toward unifying mass, gravitation, time and energy conservation. Nevertheless, it has long confronted a critical difficulty: ordinary elastic solid media necessarily support both transverse and longitudinal waves, yet no significant propagating elastic longitudinal wave in vacuum has been observed in astronomical observations. This circumstance has prevented such models from gaining broad acceptance within mainstream physics. Zou (2025) presented a spacetime model endowed with dynamical degrees of freedom including quantum spin, resonance, axial rotation, inter quanta spacing, and deformation energy modulation. Relying on a two layer architecture of a sub Planckian elastic substrate plus a Planck scale quantum network, together with the metric frequency mirror postulate, the model provides an internally consistent mechanism explaining why dynamic longitudinal deformations do not evolve into far field propagating longitudinal waves. Upon relaxation, the potential energy of longitudinal deformation is converted into elevated local resonance and spin excitations of Space Elementary Quanta (SEQ). Via secondary coupling these excitations further convert into propagating transverse waves (electromagnetic waves and gravitational transverse mode waves). This resolves the longitudinal wave puzzle while preserving the ontological merits of the elastic space ontology. Following the Clifford Wheeler "physics as geometry" research programme, the model recovers many phenomenological results from general relativity and quantum field theory, and yields a falsifiable experimental prediction: a measurable difference between the

magnetic moments of the electron and positron.

Keywords: elastic space; Space Elementary Quanta (SEQ); physics as geometry; longitudinal wave problem; metric frequency mirror; Clifford geometry; quantum gravity

Suggested pre reading: readers are advised to consult Zou (2025) before proceeding with this text[19].

Zou, Z. K. (2025). The Thermodynamic Arrow of Time in a Double-Layer Topology-Invariant Chiral Space with Geometric (GR) and Gauge (QFT) Degrees of Freedom :Time-Entropy Mapping; Mass-Gravity Duality; Metric-Frequency Mirroring. Preprints. https://doi.org/10.20944/preprints202505.0270.v12

1. Historical origins of the elastic space paradigm and the longitudinal wave puzzle

Intellectual roots of elastic space thinking trace back to Lord Kelvin's vortex atom hypothesis (1867) and W. K. Clifford's *On the Space Theory of Matter* (1870). Clifford advanced the thesis that matter itself is constituted by geometric curvature of three dimensional space, discarding the old ether dichotomy of "container space plus filling in medium". This research programme was later taken forward by Wheeler's geometrodynamics, Sakharov's induced gravity proposal, and multiple modern elastic spacetime models.

Frameworks built on the 19th century elastic solid ether ran into a sharp contradiction. Within continuum elastic solid mechanics, given volume compression stretching degrees of freedom, dynamical equations generically produce both shear transverse waves and compressional longitudinal waves. Optical observations confirm that electromagnetic waves possess purely transverse polarisations; gravitational wave detections likewise report only two transverse polarisation modes, with no observed long range propagating longitudinal modes.

Historically older elastic ether models often eliminated longitudinal waves by imposing ad hoc boundary conditions lacking underlying physical mechanisms. This constituted a persistent conceptual weakness for elastic medium spacetime proposals.

A vital distinction should be emphasised: the Michelson Morley experiment falsified a kind of ether with separate "matter space" dualism. It does **not** rule out a Clifford style ontology where space itself is an elastic entity. The longitudinal wave problem represents an intrinsic

theoretical obstacle internal to elastic space paradigms.

Many analogue gravity studies employ fluids or elastic solids in laboratory settings to mimic curved spacetime. However, analogue gravity approaches are mostly simulators; they do not claim that spacetime *is* ontologically an elastic medium, and consequently bear no obligation to resolve the longitudinal wave problem. By contrast, ontological elastic space models must answer the core question: if space behaves like an elastic solid like medium, why are longitudinal waves not observed?

Traditional continuum elastic solid attempts to suppress longitudinal wave propagation commonly adopt one approach:

Postulate a strictly zero bulk modulus (incompressible medium). This forbids any compressive deformation outright. Yet to account for mass, the paradigm requires precisely local static compressive solitons. The two requirements contradict each other.

2. Longitudinal wave suppression mechanism within Zou's (2025) SEQ two layer architecture

Zou's SEQ model adopts a two layer fibre bundle architecture:

1) **Sub Planckian elastic substrate**: bears elastic deformation potential energy and realises spatial compression, stretching and torsional deformation.
2) **Planck scale SEQ quantum network**: Space Elementary Quanta (SEQ) are topologically stable, indivisible fundamental quanta emergent from sub Planckian constituents. Each SEQ carries two generalised coordinates: resonance frequency and resonance axis vector, together with intrinsic spin synchronized with resonance. Under low energy conditions, SEQ spin torsional deformations decouple from deformations of the sub Planckian elastic substrate; coupling is restored only at high energies.

Core postulate — the metric frequency mirror postulate: spatial compression, stretching and torsional deformation do not primarily manifest as inter element displacement travelling waves. Instead, deformations directly map onto and modulate the intrinsic resonance frequency of each SEQ.

From this postulate emerges the complete energy conversion chain:

1) **Static longitudinal compression (mass soliton states) are permitted**: a Higgs like chiral locking mechanism maintains local SEQ in compressed configurations, persistently depressing local resonance frequencies. This configuration corresponds to mass. Surrounding SEQ become stretched, giving rise to the gravitational potential. This static state produces only frequency offsets and does not generate waves.

2) **Dynamic longitudinal deformation (compressive relaxation rebound)**: deformation potential energy does not generate outward propagating longitudinal displacement waves. When local deformations relax, potential energy is converted locally into SEQ resonance and spin kinetic energy, manifesting as elevated local resonance frequencies.

3) **Secondary coupling between SEQ resonance degrees of freedom and shear type deformations**: local resonant excitations further convert into shear type perturbations, which propagate outwards toward the far field as transverse waves (electromagnetic or gravitational waves).

In short: energy channels for longitudinal deformations are redirected by the metric frequency mirror principle and transverse wave mechanisms. Energy released from dynamic deformations first enters SEQ resonance degrees of freedom and remains confined to the near field. It skips the intermediate stage of propagating longitudinal displacement waves, and only secondarily radiates outwards as transverse waves.

Static locked compressive solitons (mass) can exist, yet relaxation of dynamic spatial deformations cannot escape to the far field as longitudinal waves. The scheme simultaneously satisfies both requirements: allowing local compressive deformations and their relaxation, together with the absence of far field longitudinal waves. Volume compression degrees of freedom are retained and need not be artificially eliminated.

Consistent with Chapter 14 of Zou (2025): at low energies spin torsional deformations decouple from the elastic substrate. Magnetic fields originate in SEQ spin twistors; they must propagate via resonance shear transverse excitations and cannot propagate independently.

3. Comparison with alternative elastic spacetime approaches

1) **Contrast with classical continuum elastic solids**: the model avoids patching parameters such as elastic moduli. Instead it invokes the two layer architecture plus metric frequency mirror to remodel energy conversion pathways, drawing a clear distinction between static soliton deformations and dynamic propagating perturbations.

2) **Contrast with energy gap hypotheses**: longitudinal waves are not merely forbidden above some energy threshold. The relevant dynamical channels themselves do not exist. Dynamic compressive perturbations never manifest as displacement type travelling longitudinal waves. Even in high energy events, only local SEQ resonant excitations arise, subsequently converting into transverse wave radiation.

3) **Retained core strengths of the Clifford Wheeler programme**: mass is geometric deformation of space; gravitation arises from such deformation. Global and local energy

conservation can be naturally implemented, resolving the conceptual difficulty that general relativity lacks well defined local gravitational energy density. Gravitational redshift can be interpreted as energy exchange between deformation potential energy and SEQ resonant energy. Time is not taken as a primitive four dimensional geometric dimension; proper time is directly equivalent to the local SEQ resonance frequency.

4. Open questions and falsifiable prediction

At present this physical picture remains a conceptual framework. Multiplicative entropy possesses relatively complete derivations, yet full dynamical equations still await rigorous discrete differential geometric formalisation. Outstanding points for further development:

1) Rigorous mathematical characterisation of coupling operators: the operator describing conversion from longitudinal deformation into resonant excitation, and the secondary conversion between resonance and shear transverse wave modes.
2) How a disordered polycrystalline SEQ network statistically recovers observational Lorentz covariance on scales vastly larger than the Planck length. While conceptually self consistent, this still requires formal mathematical treatment.

Nevertheless the model delivers an explicit falsifiable experimental output. Owing to the fixed chiral ground state spin of SEQ, the electron and positron should exhibit a small but measurable discrepancy in their magnetic moment magnitudes. Should next generation high precision g factor measurements rule out such a discrepancy, core assumptions of the model are falsified. If the difference is observed, experimental support would accrue for this two layer elastic spacetime picture.

5. Conclusion

The longitudinal wave puzzle constitutes a long standing theoretical obstacle for ontological elastic space proposals dating back more than a century. Zou's (2025) SEQ quantised elastic spacetime model, built upon a sub Planckian to Planck scale two layer network together with the metric frequency mirror postulate, offers an architecture level physical explanation: potential energy released from dynamic longitudinal deformations preferentially converts into local SEQ resonant excitations, which in turn couple to produce transverse mode radiation. This mechanism accounts for the observational absence of vacuum longitudinal waves.

The framework preserves the ontological attractiveness of the Clifford Wheeler "physics as geometry" programme. It unifies mass gravity geometric duality, global

energy conservation, emergent time, and the microscopic electromagnetic picture within one single three dimensional discrete network framework. That said, full mathematical formalisation and quantitative matching to standard model parameters remain tasks for future work.

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